

Laser Therapy in Lumbar Disc Herniation Report

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R version 4.5.1 (2025-06-13) was used to conduct the analysis. Visual Inspection and Shapiro-Wilk test were performed to assess normality of data to determine the appropriate statistical tests. Data not normally distributed was presented by median and interquartile range. Mann-Whitney U test and Kruskal-Wallis test were used to compare between groups of data not normally distributed. Categorical data were presented with frequencies and an appropriate Chi-squared test or Fisher exact test was used for the comparison. Significant differences were determined by Bonferroni post hoc tests. P value <0.05 was considered statistically significant.

Results

60 patients were enrolled in the study. Table 1 represents different patients' demographics.

Table 1: Demographics and Patients Characteristics

Characteristic	Overall N = 60 ¹	High Intensity N = 20 ¹	Low Intensity N = 20 ¹	Placebo N = 20 ¹	p- value ²
Gender					>0.9
Female	24 (40%)	8 (40%)	8 (40%)	8 (40%)	
Male	36 (60%)	12 (60%)	12 (60%)	12 (60%)	
Age	41 (32, 51)	38 (31, 49)	41 (33, 50)	42 (31, 53)	0.8
BMI	26.0 (25.0, 29.0)	26.5 (24.0, 31.5)	26.0 (24.0, 27.5)	28.5 (25.0, 30.0)	0.2
Marital Status					>0.9
Married	52 (87%)	17 (85%)	17 (85%)	18 (90%)	
Single	8 (13%)	3 (15%)	3 (15%)	2 (10%)	
Roland Morris Disability Questionnaire	15.0 (13.0, 16.0)	15.0 (14.0, 16.5)	14.5 (12.0, 16.0)	14.5 (13.0, 16.5)	0.8
Schober Test	16 (15, 16)	16 (15, 16)	16 (15, 17)	16 (15, 16)	0.8
Oswestry Disability Index	0.66 (0.56, 0.71)	0.66 (0.60, 0.70)	0.62 (0.53, 0.71)	0.64 (0.56, 0.72)	0.8

Characteristic	Overall N = 60 ¹	High Intensity N = 20 ¹	Low Intensity N = 20 ¹	Placebo N = 20 ¹	p- value ²
VAS Rest	6.0 (6.0, 7.0)	7.0 (6.0, 7.5)	6.0 (5.0, 7.0)	6.0 (5.5, 7.0)	0.076
VAS Activity	8 (7, 9)	9 (8, 9)	8 (7, 9)	8 (7, 9)	0.4

¹n (%); Median (Q1, Q3)

²Pearson's Chi-squared test; Kruskal-Wallis rank sum test; Fisher's exact test

All three groups demonstrated significant improvements from baseline to post-intervention across all outcome measures (p<0.001). All within-group comparisons remained significant after Bonferroni correction for multiple testing (q<0.01). (Table 2)

Table 2: Within-group changes in patient-reported outcomes and pain scores from baseline to post-intervention

Characteristic	High Intensity			Low Intensity			Placebo		
	Before N = 20 ¹	After N = 20 ¹	p- value ²	Before N = 20 ¹	After N = 20 ¹	p- value ²	Before N = 20 ¹	After N = 20 ¹	p- value ²
RMDQ	15.0 (14.0, 16.5)	4.0 (3.0, 9.0)	<0.001	14.5 (12.0, 16.0)	7.0 (3.5, 10.5)	<0.001	14.5 (13.0, 16.5)	10.0 (5.0, 13.5)	<0.001
ST	16.0 (15.0, 16.0)	19.0 (17.5, 19.5)	0.001	16.0 (15.0, 17.0)	18.0 (16.5, 18.5)	<0.001	16.0 (15.0, 16.0)	17.0 (16.0, 18.0)	0.002
OSW	0.66 (0.60, 0.70)	0.15 (0.08, 0.51)	<0.001	0.62 (0.53, 0.71)	0.38 (0.21, 0.55)	<0.001	0.64 (0.56, 0.72)	0.53 (0.23, 0.64)	0.001
VAS-Rest	7.0 (6.0, 7.5)	0.0 (0.0, 4.0)	<0.001	6.0 (5.0, 7.0)	3.0 (0.0, 5.0)	0.001	6.0 (5.5, 7.0)	5.0 (1.5, 5.5)	0.002
VAS-Activity	8.5 (8.0, 9.0)	1.0 (1.0, 6.0)	<0.001	8.0 (7.0, 9.0)	3.5 (2.0, 5.5)	0.001	8.0 (7.0, 9.0)	5.5 (2.0, 8.0)	0.002

¹Median (Q1, Q3)

²Wilcoxon signed rank test with Bonferroni correction for multiple testing

Between-group comparisons of change scores revealed significant differences across all outcome measures. The High Intensity group demonstrated significantly greater

improvements compared to the other groups, with a median RMDQ reduction of -10.0 points, ST increase of 3.00, OSW decrease of -0.41, and VAS-Activity and VAS-Rest reductions of -7.00 and -6.00, respectively. Group differences were statistically significant for RMDQ, OSW, VAS-Activity, and VAS-Rest ($p \leq 0.003$), with most remaining significant after Bonferroni correction (Table 3)

Table 3: Between-group change scores and pairwise comparisons Kruskal–Wallis with Bonferroni-adjusted post hoc tests.

Characteristic	High Intensity N = 20 ¹	Low Intensity N = 20 ¹	Placebo N = 20 ¹	p-value ²	High Intensity vs. Low Intensity	High Intensity vs. Placebo	Low Intensity vs. Placebo
RMDQ Change	-10.0 (-11.0, -6.5)	-8.0 (-9.0, -4.0)	-3.5 (-9.0, -2.0)	0.003	0.032	<0.001	0.4
ST Change	3.00 (1.00, 4.00)	2.00 (1.00, 2.00)	1.00 (1.00, 2.00)	0.090	0.14	0.033	0.7
OSW Change	-0.41 (-0.55, -0.17)	-0.28 (-0.39, -0.11)	-0.15 (-0.34, -0.07)	0.011	0.045	0.003	0.7
VAS Activity Change	-7.00 (-8.00, -2.00)	-4.50 (-6.00, -1.00)	-1.00 (-5.50, -1.00)	0.006	0.032	0.002	0.5
VAS Rest Change	-6.00 (-7.00, -2.50)	-3.00 (-5.00, -1.50)	-1.50 (-4.00, -1.00)	0.013	0.061	0.004	0.4

¹Median (Q1, Q3)

²Kruskal-Wallis rank sum test with Bonferroni correction for multiple testing

Post hoc pairwise comparisons with Bonferroni correction revealed that the High Intensity group achieved significantly greater improvements than the Placebo group across all outcome measures (RMDQ: $p_{\text{adj}} < 0.001$; OSW: $p_{\text{adj}} = 0.003$; ST: $p_{\text{adj}} = 0.033$; VAS-Activity: $p_{\text{adj}} = 0.002$; VAS-Rest: $p_{\text{adj}} = 0.004$). The High Intensity group also demonstrated significantly superior outcomes compared to the Low Intensity group for RMDQ ($p_{\text{adj}} = 0.032$), OSW ($p_{\text{adj}} = 0.045$), and VAS-Activity ($p_{\text{adj}} = 0.032$), though differences in ST and VAS-Rest did not reach significance after correction ($p_{\text{adj}} = 0.138$ and 0.061 , respectively). In contrast, the Low Intensity group did not differ significantly from the Placebo group on any outcome measure after Bonferroni correction (all $p_{\text{adj}} > 0.42$), suggesting that low-intensity intervention provided minimal additional benefit beyond placebo effects.

Figure 1 and 2 illustrates the median and IQR values of all outcome measures before and after intervention for the High Intensity, Low Intensity, and Placebo groups. All three groups showed improvements across disability scores. The visual representation highlights the dose-dependent relationship between intervention intensity and clinical outcomes, with greater improvements observed in functional capacity and pain reduction as intervention intensity increased.

Figure 1: Faceted median line plot (IQR error bars) of pre-post outcomes by treatment group

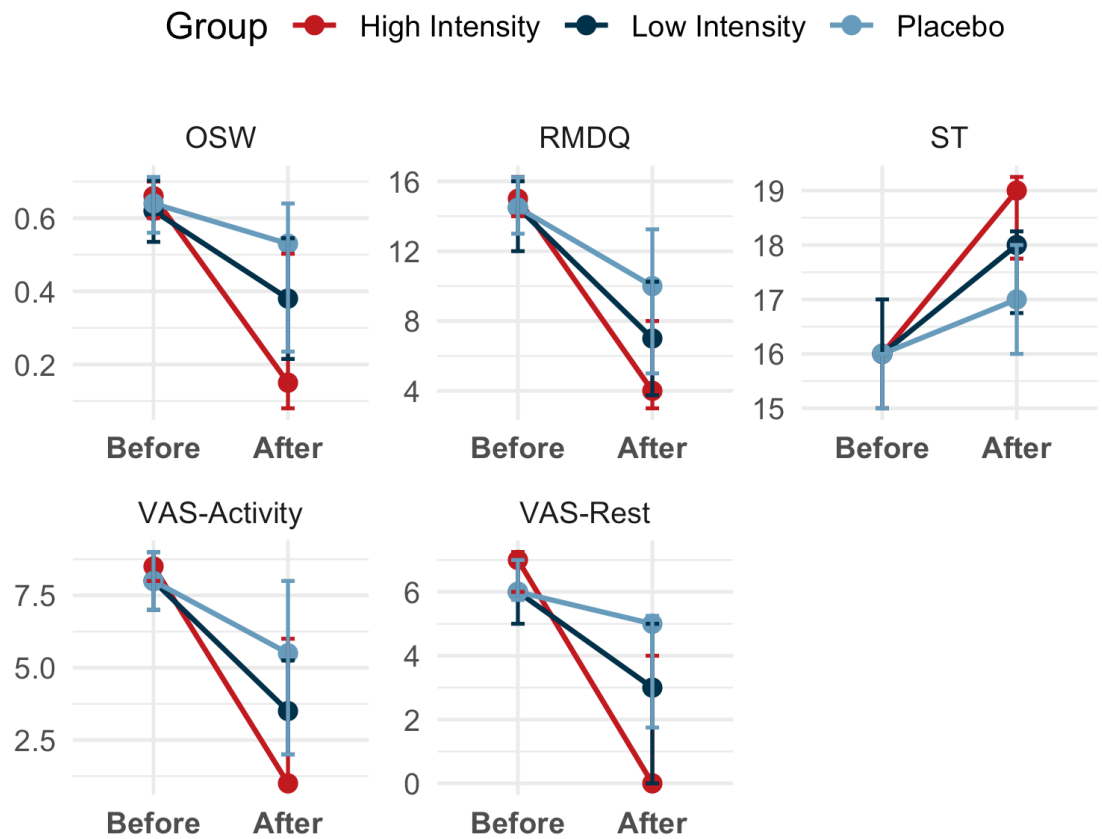


Figure 2: Comparison of Pre- and Post-Intervention Outcome Measures Across Treatment Groups

